Practice workbook · Break-even output using contribution

Finding break-even level of output as fixed costs divided by contribution per unit (price minus variable cost per unit).

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: BUS-CALC-06

1. Question 4 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

4 *PPS* is planning to introduce a new line of dog treats. **Figure 4** shows some financial information for this product.

	€
Monthly fixed costs	1 000
Variable cost per unit	3.25
Selling price	4.50

Figure 4

(a) Calculate the break-even level of output for this new line of dog treats. You are advised to show your working.

(2)

..... dog treats

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1. Question 4 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

(b) Analyse the importance of cash to *PPS*.

(6)

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1. Question 4 · part a · 2 marks · Working and answer

Working and answer

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2. Question 3 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

3 *Backyard Shoez* is analysing its financial position before it begins selling shoes to men and children.

(a) Define the term **profit**.

(1)

(b) Outline **one** impact on *Backyard Shoez* if interest rates increase.

(2)

Backyard Shoez's monthly fixed costs are 100 000 Ks with a variable cost of 3 500 Ks for a pair of shoes, at a selling price of 4 300 Ks.

(c) Calculate the break-even point. You are advised to show your working.

(2)

Ks

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2. Question 3 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

(d) Analyse the benefits to *Backyard Shoez* of using break-even analysis to plan the launch of its new shoes for men and children.

(6)

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2. Question 3 · part c · 2 marks · Working and answer

Working and answer

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3. Question 4 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

4 (a) Calculate the number of people needed to break-even for this course. You are advised to show your workings.

(2)

..... people

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3. Question 4 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

(b) Analyse the benefits for *Loaf* of employing casual staff.

(6)

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3. Question 4 · part a · 2 marks · Working and answer

Working and answer

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4. Question 3 · part c · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

3 *HRMW* employs staff to design the products it manufactures. The aim is to add features so that the products are better than any others on the market.

(a) Define the term **e-tailers**.

(1)

(b) Outline **one** reason why *HRMW* uses robotics in the production of its goods.

(2)

If *HRMW* produce tennis racquets with the following costs:

Fixed costs = 300 000 Rupees

Variable costs = 3 000 Rupees

Sale price = 5 000 Rupees

(c) Calculate the number of tennis racquets that would have to be made to break-even. You are advised to show your working.

(2)

..... Tennis racquets

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4. Question 3 · part c · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

(d) Analyse the limitations for *HRMW* of using break-even charts when planning the manufacture of tennis racquets.

(6)

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4. Question 3 · part c · 2 marks · Working and answer

Working and answer

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5. Question 3 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

3 (a) Define the term **remuneration**.

(1)

(b) Outline **one** reason why *NOF* would choose to use seasonal staff when harvesting its products.

(2)

NOF is looking to expand its product range by making and selling tubs of ice cream.

A tub of ice cream will sell for £2.50.

Variable costs will be £1.10 per tub of ice cream with fixed costs of £77 per day.

(c) Calculate the number of tubs of ice cream *NOF* will have to sell each day to break-even. You are advised to show your working.

(2)

.....Tubs of ice cream

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5. Question 3 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

(d) Analyse how useful break-even calculations can be for *NOF* when deciding whether to include ice cream in its product range.

(6)

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Turn over

5. Question 3 · part c · 2 marks · Working and answer

Working and answer

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6. Question 1 · part e · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

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1 *Aude Rose* is a flower shop in Paris, France. It was set up by Aude Anglaret after she graduated with a qualification in floristry. It opened a shop in 2009 in the outskirts of Paris. Since then it has gone from strength to strength increasing its market share. To meet this increase in demand it has closed its first shop and opened a larger one in the centre of Paris with a higher rent and a possible increase in customers. It currently employs four florists. *Aude Rose* has been recognised for its high-quality flower arrangements that have been seen in several wedding magazines.

(a) (i) Which **one** of the following describes a minimum wage?

Select **one** answer.

(1)

- ☒ **A** Fixed amount per year and paid monthly
- ☒ **B** The lowest amount a business can legally pay its employees
- ☒ **C** Payment made to an employee for working additional hours
- ☒ **D** Payment made to an employee for reaching a target

(ii) Which **one** of the following describes the cash cow category in the Boston matrix?

Select **one** answer.

(1)

- ☒ **A** High market growth, high market share
- ☒ **B** High market growth, low market share
- ☒ **C** Low market growth, high market share
- ☒ **D** Low market growth, low market share

(iii) Which **one** of the following would give a business a negative cash flow?

Select **one** answer.

(1)

- ☒ **A** Cash inflow is greater than cash outflow
- ☒ **B** Costs are greater than profits
- ☒ **C** Profits are greater than costs
- ☒ **D** Cash outflow is greater than cash inflow

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6. Question 1 · part e · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

(iv) Which **one** of the following is a recruitment document that outlines the tasks and responsibilities of a job?

Select **one** answer.

(1)

- ☒ **A** Job description
- ☒ **B** Person specification
- ☒ **C** Application form
- ☒ **D** Curriculum vitae

To wrap its flowers *Aude Rose* buys 150 metres of wrapping paper from London, England every week at a cost of £0.46 per metre.

In October 2022 the exchange rate was £1 = €1.15

(v) Which **one** of the following is the total cost, in euros, of buying a four week supply of wrapping paper?

Select **one** answer.

(1)

- ☒ **A** €79.35
- ☒ **B** €276.00
- ☒ **C** €317.40
- ☒ **D** €396.75

Aude Rose provides flowers for large events. The most popular is a selection of traditional flowers selling for €65. The cost of making this selection of flowers is €42.

(vi) Which **one** of the following is the markup for a selection of traditional flowers?

Select **one** answer.

(1)

- ☒ **A** 54.76%
- ☒ **B** 64.62%
- ☒ **C** 182.61%
- ☒ **D** 282.61%

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6. Question 1 · part e · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

(b) Define the term **economies of scale**.

(1)

(c) Define the term **stakeholder**.

(1)

(d) State **one** benefit to *Aude Rose* of using just-in-time (JIT) production.

(1)

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Aude Rose has fixed costs of €50 000 a year with a selling price of €50 and a variable cost of €10 per bunch of flowers sold.

(e) Calculate how many bunches of flowers *Aude Rose* will have to sell to break-even. You are advised to show your working.

(2)

..... bunches of flowers

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Turn over

6. Question 1 · part e · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

(f) Explain **one** reason why a business might fail.

(3)

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6. Question 1 · part e · 2 marks · Working and answer

Working and answer

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7. Question 1 · part a.vi · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part a.vi only.

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1 *Irsi Chocolatier* is a chocolate shop in Brussels, the capital city of Belgium. There are many other chocolate shops nearby. Opened in 1989 by the Corne family the business now has four directors and three employees. Florent Corne is the Managing Director with three other family members as Directors. *Irsi Chocolatier* opens from Tuesday till Saturday. The shop sells a variety of handmade chocolates and jelly fruit sweets. It delivers locally and has a website for information purposes only.

(a) (i) Which **one** of the following functions is responsible for public relations?

Select **one** answer.

(1)

- ☒ **A** Human resources
- ☒ **B** Finance
- ☒ **C** Marketing
- ☒ **D** Production

(ii) Which **one** of the following is the difference between cash inflow and cash outflow?

Select **one** answer.

(1)

- ☒ **A** Opening balance
- ☒ **B** Net cash flow
- ☒ **C** Closing balance
- ☒ **D** Cash flow forecast

(iii) Which **one** of the following is the sector that converts raw materials into products?

Select **one** answer.

(1)

- ☒ **A** Public
- ☒ **B** Tertiary
- ☒ **C** Primary
- ☒ **D** Secondary

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7. Question 1 · part a.vi · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part a.vi only.

(iv) Which **one** of the following is an example of a good?

Select **one** answer.

(1)

- ☒ **A** Motor vehicle
- ☒ **B** Financial advice
- ☒ **C** Car washing
- ☒ **D** Gardening

In 2021 *Irsi Chocolatier*'s revenue was €311 625. Each year it has seen its revenue increase.

(v) Which **one** of the following would be a 3% increase in revenue for *Irsi Chocolatier*?

Select **one** answer.

(1)

- ☒ **A** €317 857.50
- ☒ **B** €319 415.63
- ☒ **C** €320 973.75
- ☒ **D** €322 531.88

Irsi Chocolatier's monthly fixed costs are €950 and variable costs per luxury box of chocolates are €2.60.

(vi) Which **one** of the following is the break-even point when a box of luxury chocolates is priced at €15.10?

Select **one** answer.

(1)

- ☒ **A** 75
- ☒ **B** 76
- ☒ **C** 77
- ☒ **D** 78

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7. Question 1 · part a.vi · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part a.vi only.

(b) Define the term **limited liability**.

(1)

(c) Define the term **social enterprise**.

(1)

(d) State **one** way *Irsi Chocolatier* could use the Boston matrix.

(1)

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A customer buying five boxes of chocolates for a total of €75.50 is given a 6% discount.

(e) Calculate, to two decimal places, the cost of these chocolates after the discount has been applied. You are advised to show your working.

(2)

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7. Question 1 · part a.vi · 1 mark · Working and answer

Working and answer

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8. Question 1 · part a.vi · 1 mark

2023 NOV · Paper 01

Further practice · Attempt part a.vi only.

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1 *As We Grow* (*AWG*) is a clothing business in Iceland. The social objective of the business is to use natural materials, such as wool, and produce environmentally friendly products. It aims to design clothes that will last a long time, to encourage families to buy fewer clothes and reduce waste in society.

AWG's increasing product range now includes jumpers, dresses, shirts and baby clothes. These are produced in small workshops that employ family and friends. Waste is kept to a minimum and any leftover material is used to make scarves and hats. The business has one store in Iceland. It uses its website to advertise and sell its clothing online to customers in other countries. The business has received the Icelandic Design Award for its contribution to the environment.

(a) (i) Which **one** of the following is shown on the statement of financial position?

Select **one** answer.

(1)

- ☒ **A** Sales
- ☒ **B** Cost of sales
- ☒ **C** Current assets
- ☒ **D** Gross profit

(ii) Which **one** of the following is an element of the marketing mix?

Select **one** answer.

(1)

- ☒ **A** Price
- ☒ **B** Partnership
- ☒ **C** Production
- ☒ **D** Profit

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8. Question 1 · part a.vi · 1 mark

2023 NOV · Paper 01

Further practice · Attempt part a.vi only.

A business has fixed costs of £6 000 a month. Variable costs are £5 per unit and 8,000 units are made each month.

(iii) Which **one** of the following is the total monthly cost?

Select **one** answer.

(1)

- ☒ **A** £ 6 005
- ☒ **B** £14 000
- ☒ **C** £30 000
- ☒ **D** £46 000

(iv) Which **one** of the following job roles is most likely to be associated with capital-intensive production?

Select **one** answer.

(1)

- ☒ **A** Hairdresser
- ☒ **B** Car assembly worker
- ☒ **C** Soft fruit picker
- ☒ **D** Chef

(v) Which **one** of the following is an internal source of finance?

Select **one** answer.

(1)

- ☒ **A** Overdraft
- ☒ **B** Venture capital
- ☒ **C** Retained profit
- ☒ **D** Crowdfunding

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8. Question 1 · part a.vi · 1 mark

2023 NOV · Paper 01

Further practice · Attempt part a.vi only.

The currency of Iceland is the krona (ISK).

AWG's weekly fixed costs are 440 000 ISK and the variable cost per jumper is 2 500 ISK. Each jumper is sold for 8 000 ISK.

(vi) Which **one** of the following is the weekly break-even point?

Select **one** answer.

(1)

- ☒ **A** 8
- ☒ **B** 55
- ☒ **C** 80
- ☒ **D** 176

(b) Define the term **profit**.

(1)

(c) Define the term **span of control**.

(1)

(d) State **one** likely variable cost for *AWG*.

(1)

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8. Question 1 · part a.vi · 1 mark · Working and answer

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9. Question 1 · part a.v · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part a.v only.

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1 Gianni Polvani opened his first gym, *Four Fitness*, over 20 years ago. He now employs 14 highly-qualified gym instructors in his two gyms. Both gyms are in Italy, one in Arezzo and the other in Castiglion Fiorentino.

The gym encourages its members to take part in all the physical activities it offers. Members can take part in a number of social activities as well.

Four Fitness has a website which is primarily for information only. The website details such things as opening hours and descriptions of available activities. It cannot be used to book activities or access personal training plans.

(a) (i) Which **one** of the following is a debt payable within 12 months?

Select **one** answer.

(1)

- ☒ **A** Current asset
- ☒ **B** Current liability
- ☒ **C** Non-current asset
- ☒ **D** Non-current liability

(ii) Which **one** of the following is a pricing strategy where a business sets an initial high price for a product new to the market?

Select **one** answer.

(1)

- ☒ **A** Skimming
- ☒ **B** Competition
- ☒ **C** Penetration
- ☒ **D** Cost plus

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9. Question 1 · part a.v · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part a.v only.

(iii) Which **one** of the following is a role of the human resources department?

Select **one** answer.

(1)

- ☒ **A** Reducing costs
- ☒ **B** Designing products
- ☒ **C** Customer service
- ☒ **D** Recruiting employees

(iv) Which **one** of the following is a benefit to a shareholder of having limited liability?

Select **one** answer.

(1)

- ☒ **A** Choosing who can buy shares
- ☒ **B** May build a better relationship with suppliers
- ☒ **C** Own possessions are not at risk
- ☒ **D** Employee motivation will increase

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9. Question 1 · part a.v · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part a.v only.

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Four Fitness is considering introducing sports drinks to sell to its customers and to do this it needs to hire a fridge to store the sports drinks. To decide whether it is a good business decision, it needs to work out the break-even level of output.

Figure 1 shows some financial data to assist this decision.

	€
Fixed costs	280
Variable cost per unit	0.70
Revenue per unit	1.50

Figure 1

(v) Which **one** of the following is the break-even level of output?

Select **one** answer. (1)

- ☒ **A** 280
- ☒ **B** 350
- ☒ **C** 420
- ☒ **D** 616

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9. Question 1 · part a.v · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part a.v only.

Figure 2 is a financial extract from a gym equipment business that supplies *Four Fitness*.

	€
Total revenue	132 165
Total labour costs	54 000
Total material costs	17 000
Other costs	17 400

Figure 2

(vi) Which **one** of the following statements from **Figure 2** correctly applies to the gym equipment business?

Select **one** answer. (1)

- ☒ **A** It made a loss of €88 400
- ☒ **B** It made a loss of €61 165
- ☒ **C** It made a profit of €43 765
- ☒ **D** It made a profit of €78 165

(b) Define the term **trade bloc**. (1)

(c) Define the term **job production**. (1)

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9. Question 1 · part a.v · 1 mark · Working and answer

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